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Genel

Duyurular

LAB

Lab 2

> 5 December Lab

> 11 December Lab

> 25 December Final Exam

CSE101L / Genel / Lab 2



Description

Submission view

📅 Available from: Perşembe, 28 Kasım 2024, 9:00 AM
📅 Due date: Perşembe, 28 Kasım 2024, 12:10 PM
📁 Requested files: Lab.java ( Download)
Type of work:  Individual work

In this lab, you will implement three methods within a class named Lab. You are provided with the method names, parameter types, and what each method should return. Your task is to write each method according to the given descriptions and ensure they produce the correct output based on the sample inputs provided. Note that parameter names are flexible and can be chosen by you.

Question 1: Method Name: factorial

- Parameters:**
 - int (an integer value)
- Returns:** A String representing the factorial calculation steps and result.

Description:
The factorial method calculates the factorial of a non-negative integer. It returns a string that shows the multiplication sequence leading to the factorial result. If the input integer is negative, the method should return "Undefined". For input values 0 or 1, the method should return "1".

- Examples:**
 - Input: 5
 - Output: "1 * 2 * 3 * 4 * 5 = 120"

- Input: -3
- Output: "Undefined"

- Input: 1
- Output: "1"

Question 2: Method Name: maximumProduct

- Parameters:**
 - int[] (an array of integers)
- Returns:** An int representing the maximum product of any three numbers from the array.

Description:
The maximumProduct method takes an array of integers and computes the maximum product that can be obtained by multiplying any three numbers from the array. The array may contain both positive and negative integers. The method should consider all possible combinations to find the maximum product.

- Examples:**
 - Input: [1, 2, 3]
 - Output: 6
- Input: [-10, -10, 5, 2]
- Output: 500
- Explanation: The maximum product is -10 * -10 * 5 = 500.
- Input: [0, -1, 3, 100, 70, 50]
- Output: 350000
- Explanation: The maximum product is 100 * 70 * 50 = 350000.

Question 3: Method Name: saleCar

- Parameters:**
 - String[] (an array of car model names)
 - String (the car model to be sold)
- Returns:** A String[] containing the remaining car models after removal.
- Description:

The saleCar method receives an array of car models and a specific car model to sell. It returns a new array containing the remaining car models after removing all occurrences of the specified car model. The comparison should be case-insensitive, so "Toyota" and "toyota" are considered the same.

- Examples:**
 - Input:
 - Car models: ["Toyota", "BMW", "Audi", "toyota", "Mercedes"]
 - Car to sell: "Toyota"
 - Output: ["BMW", "Audi", "Mercedes"]
 - Input:
 - Car models: ["Ford", "Chevrolet", "Ford", "Honda"]
 - Car to sell: "ford"
 - Output: ["Chevrolet", "Honda"]
 - Input:
 - Car models: ["Nissan", "Hyundai", "Kia"]
 - Car to sell: "BMW"
 - Output: ["Nissan", "Hyundai", "Kia"]
 - Explanation: Since "BMW" is not in the list, the original array remains unchanged.